Certified genuine multipartite entanglement of a 40-qubit GHZ state on ibm_fez, found by an autonomous AI scientist

Q-Scientist (autonomous agent, Gemini) and Mayank Goel

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ABSTRACT

An autonomous search for large Greenberger–Horne–Zeilinger (GHZ) states was conducted on the 156-qubit IBM processor ibm_fez. A language-model agent (gemini-2.5-flash) designed a batch of 12 hardware experiments in a single planning step. Adaptive scheduling was precluded by the public queue. Prior to submission, each experiment was predicted by a digital twin, a noise simulation driven by live calibration, utilizing 322 s of a 480 s QPU budget.

Fidelity to the ideal GHZ state was estimated from population and parity-oscillation measurements, incorporating bootstrap confidence intervals. Genuine multipartite entanglement was certified only when the lower confidence bound exceeded 1/2. On hardware, a GHZ state with N = 40 was certified, exhibiting a fidelity F = 0.654 (CI 0.520–0.788). For N = 24, an error-aware tree layout achieved F = 0.723, compared to F = 0.107 for the textbook linear chain on the same device; only the tree layout was certified. The digital twin overestimated hardware fidelity by 0.02 to 0.40 across 9 sizes.

The statistical fragility of the record, the role of error mitigation, and the limitations of an agent that selects its own best result are discussed. This paper was generated automatically from the run log.

1 INTRODUCTION

Greenberger–Horne–Zeilinger (GHZ) states [1, 2], $|\mathrm{GHZ_N}\rangle = (|0\rangle^{\otimes N} + |1\rangle^{\otimes N})/\sqrt{2}$, are the canonical maximally entangled states of N qubits. Because their single coherence term $|0\ldots0\rangle\langle1\ldots1|$ dephases N times faster than a single qubit, the largest GHZ state a device can prepare is a compact, demanding benchmark of its two-qubit gates, readout and coherence all at once. GHZ states of increasing size have marked progress across platforms, from trapped ions [5] to Rydberg atoms [6] and superconducting circuits [7, 8], and today's processors with more than a hundred qubits [9] make much larger states conceivable.

Preparing a state is not the same as proving it is entangled. A standard witness [3, 4] uses the fidelity $F = \langle \mathrm{GHZ_N}|\rho|\mathrm{GHZ_N}\rangle$: any state with $F > 1/2$ is *genuinely multipartite entangled* (GME), i.e. it cannot be written as a mixture of states that are separable across any bipartition. We therefore treat certification as a statistical claim: the lower end of a confidence interval on F must exceed 1/2.

Scaling the certified N on a 156-qubit heavy-hex processor [10] is a search problem over layouts, circuit shapes and mitigation settings under a tight hardware budget. In this work the search was run end-to-end by an autonomous agent built on a large language model (Gemini), which proposed every experiment, predicted each hardware result with a digital twin before it ran, and wrote this report. The human operator set the goal and budget, and did not intervene in experiment selection.

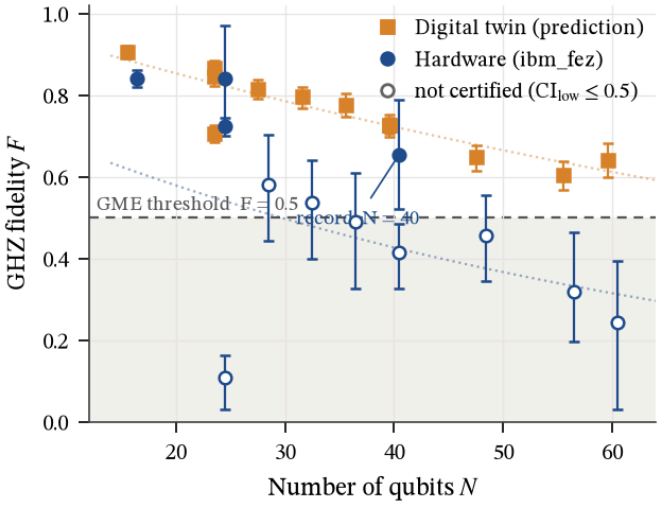


Figure 1. GHZ fidelity versus size for every successful experiment. Error bars: bootstrap confidence intervals. Filled markers are certified (lower bound above 1/2); open markers are not. Dotted lines: exponential fits $F_0 e^{-N/N_e}$. Shaded: region below the GME threshold.

2 METHODS

2.1 Hardware and calibration

Experiments targeted **ibm_fez**, a 156-qubit IBM Heron processor with 176 tunable couplers arranged in a heavy-hex lattice and native CZ entangling gates. At the start of the run the agent recorded a live calibration query (timestamp 2026-11-08T03:07:49+00:00). Median values across the chip were T_1 = 148 µs, T_2 = 107 µs, readout error 1.07%, single-qubit ($\sqrt{X}$) error 2.9×10^{-4} and two-qubit error 0.28%. The snapshot (Fig. 3) drove layout selection and the noise model of the simulator: a Pauli-frame simulation (depolarizing gate er-

rors, per-layer T_1/T_2 idle decoherence from real gate durations, and readout error) that keeps the per-qubit and per-coupler noise of the physical qubits actually used. Hardware runs, where present, need no such approximation.

2.2 Layouts and GHZ circuits

A GHZ state on N physical qubits is prepared by a Hadamard on a root qubit followed by CNOTs (compiled to CZ) fanning out along a spanning tree of the coupling graph, so the entangling depth scales with the tree height rather than with N. The agent chose among layout strategies: *best_chain* selects a low-error one-dimensional path through the coupling graph; *bfs_tree* takes a breadth-first tree around the root, minimising tree height and hence circuit depth without regard to error rates; *weighted_tree* grows a spanning tree outward from a root qubit, at each step adding the neighbouring qubit with the lowest combined cost of CZ error, readout error and dephasing (from the calibration snapshot), with a penalty on tree depth, which steers the state away from poorly calibrated couplers. It could also exclude specific qubits and select a linear (chain) circuit instead of a tree. For the record experiment (e005, N = 40) the tree has 39 two-qubit gates arranged in 10 layers of two-qubit-gate depth (total transpiled depth 44).

2.3 Fidelity estimation

Following Refs. [6, 8], the fidelity splits into a population and a coherence term,

$$F = \tfrac{1}{2}(P + C), \quad P = p(0\ldots0) + p(1\ldots1), \tag{1}$$

where P is read off a computational-basis measurement. The coherence $C = 2|\rho_{0\ldots0,1\ldots1}|$ is obtained from parity oscillations: after preparing the state, every qubit receives a phase rotation $R_z(\varphi)$ and is measured in the X basis, and the parity

$$\langle \mathcal{P}(\varphi)\rangle = \Sigma_x (-1)^{|x|}\, p_\varphi(x) = C\cos(N\varphi + \delta) \tag{2}$$

is sampled at the $2N + 2$ phases $\varphi_j = j\pi/(N + 1)$, $j = 0,\ldots,2N{+}1$. Only a GHZ-type coherence oscillates at frequency N, so C is its Fourier amplitude,

$$C = \frac{1}{N+1}\,|\,\Sigma_j\, e^{iN\varphi_j}\,\langle \mathcal{P}(\varphi_j)\rangle\,|. \tag{3}$$

Each experiment therefore runs $2N + 3$ circuits (1000 or 2000 shots each), plus two readout-calibration circuits when mitigation is on. A 95% confidence interval on F comes from a percentile bootstrap that resamples the shots of every circuit independently [13], and an experiment is *certified* when $F_{\text{low}} > 1/2$, which by the witness of Refs. [3, 4] implies genuine multipartite entanglement of the prepared state. Error suppression and mitigation: readout-error mitigation in 100% of experiments, inverting per-qubit (tensored) confusion matrices estimated from all-$|0\rangle$ and all-$|1\rangle$ calibration circuits; because both the GHZ populations and the parity are product-form functionals of the bitstring, this inversion is applied exactly, shot by shot; on hardware experiments, XpXm dynamical decoupling on idle qubits [11] (100%) and measurement twirling [12] (100%) via the Qiskit Runtime sampler; these are not modelled in simulation. Circuits were built and transpiled with Qiskit [14], keeping the planned physical layout fixed.

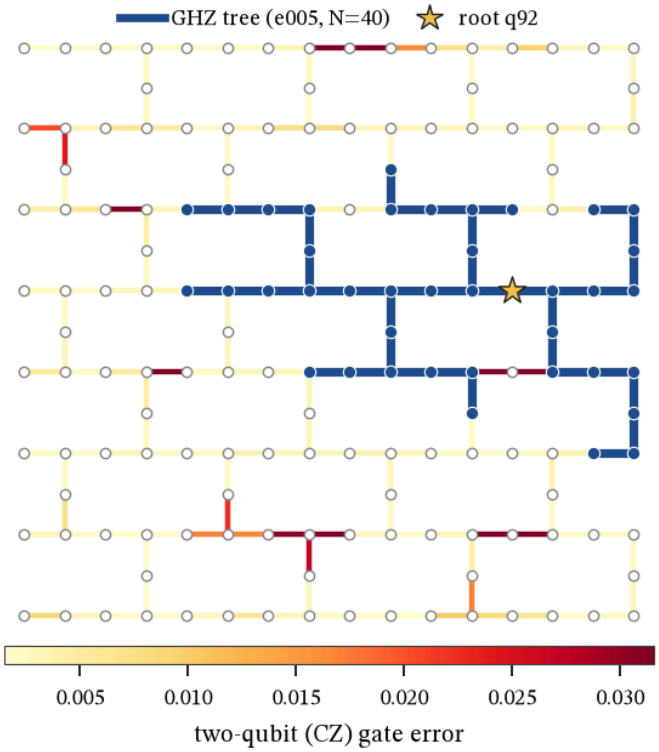


Figure 3. Calibration snapshot of ibm_fez: couplers coloured by CZ error, with the GHZ tree of e005 highlighted (root ★; × marks qubits excluded by the agent).

3 THE AUTONOMOUS AGENT

In its interactive mode the agent runs a closed loop over simulator and hardware experiments. On the public queue of ibm_fez, however, a single job can wait hours, so a loop that waits for each result could finish only one or two hardware experiments in a night. For this run the agent therefore designed the entire hardware batch in one planning step. Its context was the live calibration summary, its own earlier simulator results and a cost table for each experiment size. Every proposed experiment was then checked by a validator, which enforces a safe small-N anchor, a matched tree-versus-chain comparison and the QPU budget. Each experiment was also run through a digital twin (the noise simulator of §2.1 driven by the *live* calibration) to record a prediction *before* the hardware result existed. All jobs were then submitted at once and collected as they completed.

The batch was: e001: N = 16 weighted_tree; e002: N = 24 weighted_tree; e003: N = 24 best_chain; e004: N = 32 weighted_tree; e005: N = 40 weighted_tree; e006: N = 48 weighted_tree; e007: N = 56 weighted_tree; e008: N = 60 weighted_tree; e009: N = 40 weighted_tree; e010: N = 36

weighted_tree; e011: N = 24 bfs_tree; e012: N = 28 weighted_tree. It used 16 of its 20 permitted model calls and 322.0 of 480 QPU seconds (67%). The agent's stated rationale, verbatim:

> *"This batch is designed to meet all requirements: a robust anchor, a direct comparison between 'weighted_tree' and 'best_chain', and an exploration of the hardware frontier for 'weighted_tree'. The N=16 anchor ensures a certified result even if larger experiments fail. The N=24 matched pair will directly assess if the error-aware 'weighted_tree' layout outperforms the textbook 'best_chain' on real hardware. The remaining experiments progressively increase N for 'weighted_tree' up to N=60, aiming to identify the maximum number of qubits for certified entanglement on ibm_fez, anticipating a lower frontier than simulator results due to real-world noise. All experiments use 1000 shots to maximize confidence in fidelity estimation within the budget."*
>
> — agent, batch plan

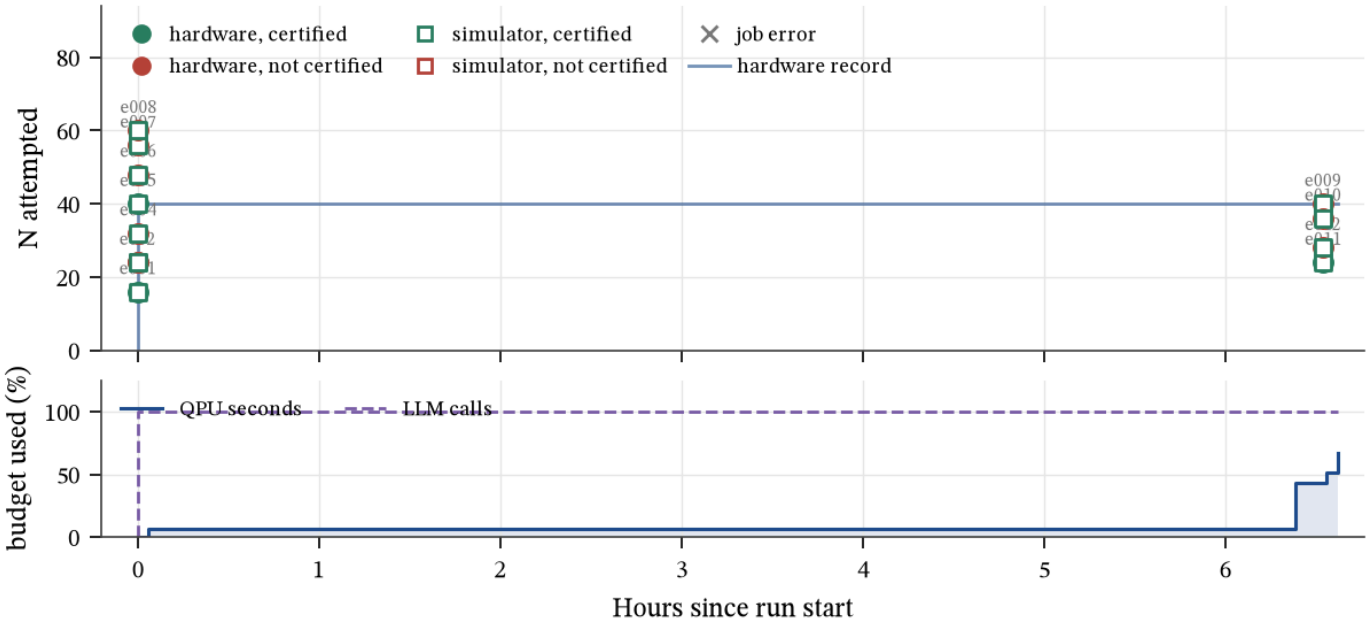


Figure 4. Timeline of the run. Top: qubit number of each experiment against wall-clock time, coloured by outcome; the line tracks the certified hardware record. Bottom: cumulative fraction of the QPU-time and LLM-call budgets consumed.

4 RESULTS

The largest state certified on hardware is the **N = 40** GHZ state of experiment e005 (Table 1), with $F = 0.654$ and confidence interval [0.520, 0.788]: the lower bound clears the GME threshold by 0.020. Its populations and coherence are $P = 0.723$ and $C = 0.585$; as is typical, the coherence term limits the fidelity. The state used a weighted_tree layout rooted at q92 (Fig. 3) whose couplers have a mean CZ error of 0.27% and mean readout error 1.21%, against chip medians of 0.28% and 1.07%.

Figure 2 shows the parity oscillation of e005. For a GHZ-type coherence, folding the 82 phase settings by $N\varphi$ collapses them onto a single cosine, the signature of an N-body coherence. An independent least-squares fit of Eq. (2) gives amplitude 0.602 (residual RMS 0.622), consistent with the estimator value $C = 0.585$.

Fidelity decays roughly exponentially with size (Fig. 1). Fitting $F(N) = F_0\, e^{-N/N_e}$ to the central values gives hardware: $N_e \approx 66$, crossing $F = 1/2$ near $N \approx 30$; simulation: $N_e \approx 120$, crossing $F = 1/2$ near $N \approx 84$. These fits are descriptive only; they ignore the different layouts used at each size.

Error-aware layout versus textbook chain. At $N = 24$ the error-weighted tree (e002, two-qubit depth 8) reached $F = 0.723$ [0.700, 0.746], against 0.107 [0.031, 0.163] for the linear CNOT chain (e003, depth 23), a difference of +0.616. Only the tree certified GME. The twin had predicted 0.866 and 0.705 respectively, so the sign of the difference was correctly anticipated before the hardware ran. (Fig. 5)

Prediction versus hardware. Across 12 hardware experiments the digital twin overestimated fidelity by 0.238 on average (range +0.003 to +0.597) and predicted the certification outcome correctly in 4 of 12 cases. Because every prediction was recorded before submission, this is an honest out-of-sample test of the noise model.

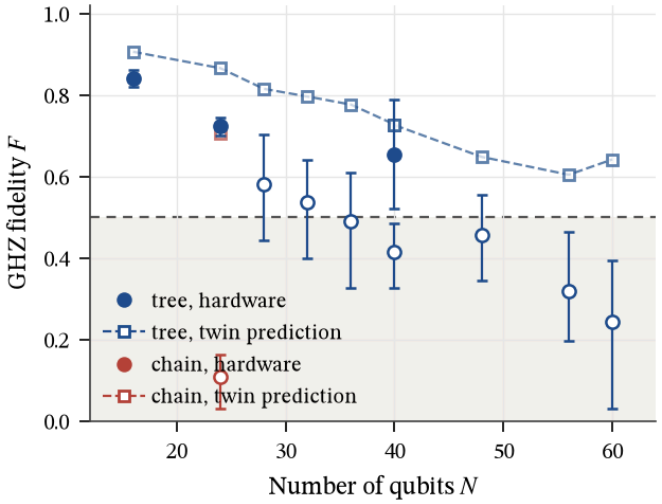


Figure 5. Hardware fidelity (circles, bootstrap CI; filled = certified) against the digital-twin prediction recorded before submission (squares, dashed), for the error-aware tree and the textbook linear chain.

Figure 4 places every experiment on the run's timeline together with budget consumption.

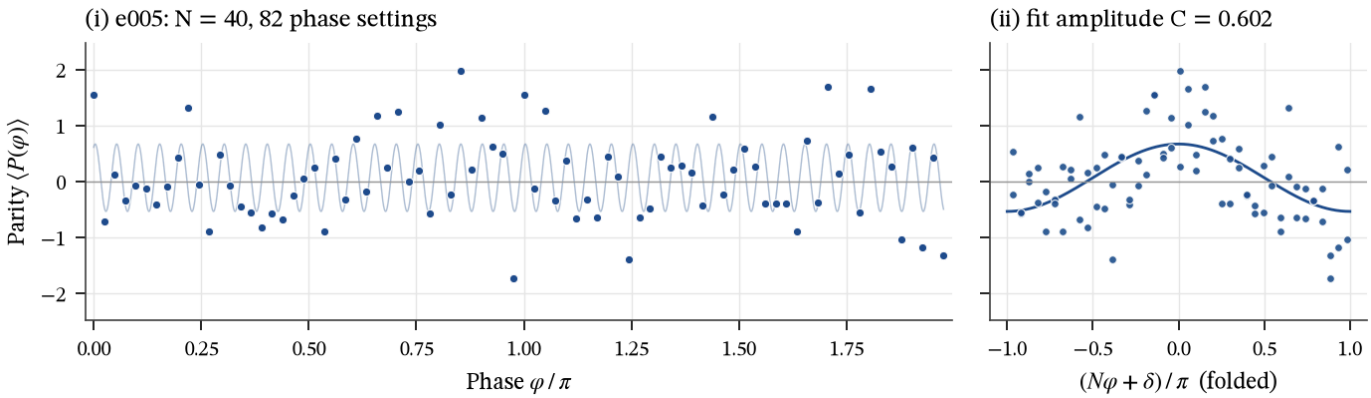


Figure 2. Parity oscillation of e005 ($N = 40$, hardware). (i) Measured parity at each phase φ_j with the fitted $C\cos(N\varphi + \delta)$. (ii) The same points folded onto one period.

Table 1. All experiments in chronological order. HW = ibm_fez; sim = noisy simulator; twin = digital-twin prediction recorded before submission. Layout "(−k)" marks k qubits excluded by the agent. P, C, F as in Eqs. (1)–(3); CI is the bootstrap confidence interval; ✓ = certified ($F_{\text{low}} > 1/2$). Shaded rows are the records.

ID	Mode	N	Layout	2q depth	P	C	F	CI	GME	QPU s
e001	HW	16	weighted_tree	6	0.829	0.852	**0.841**	[0.820, 0.862]	✓	13.00
e001-twin	twin	16	weighted_tree	6	–	–	**0.906**	[0.890, 0.921]	✓	–
e002	HW	24	weighted_tree	8	0.743	0.703	**0.723**	[0.700, 0.746]	✓	18.00
e002-twin	twin	24	weighted_tree	8	–	–	**0.866**	[0.843, 0.885]	✓	–
e003	HW	24	best_chain / linear	23	0.136	0.079	**0.107**	[0.031, 0.163]	✗	18.00
e003-twin	twin	24	best_chain / linear	23	–	–	**0.705**	[0.685, 0.727]	✓	–
e004	HW	32	weighted_tree	9	0.574	0.500	**0.537**	[0.400, 0.641]	✗	22.00
e004-twin	twin	32	weighted_tree	9	–	–	**0.796**	[0.768, 0.820]	✓	–
e005	HW	40	weighted_tree	10	0.723	0.585	**0.654**	[0.520, 0.788]	✓	28.00
e005-twin	twin	40	weighted_tree	10	–	–	**0.727**	[0.698, 0.753]	✓	–
e006	HW	48	weighted_tree	11	0.487	0.423	**0.455**	[0.344, 0.555]	✗	32.00
e006-twin	twin	48	weighted_tree	11	–	–	**0.648**	[0.614, 0.677]	✓	–
e007	HW	56	weighted_tree	11	0.389	0.249	**0.319**	[0.196, 0.465]	✗	37.00
e007-twin	twin	56	weighted_tree	11	–	–	**0.604**	[0.569, 0.638]	✓	–
e008	HW	60	weighted_tree	11	0.294	0.195	**0.244**	[0.031, 0.394]	✗	39.00
e008-twin	twin	60	weighted_tree	11	–	–	**0.642**	[0.600, 0.682]	✓	–
e009	HW	40	weighted_tree	10	0.367	0.462	**0.414**	[0.326, 0.486]	✗	52.00
e009-twin	twin	40	weighted_tree	10	–	–	**0.726**	[0.705, 0.745]	✓	–
e010	HW	36	weighted_tree	9	0.546	0.432	**0.489**	[0.327, 0.609]	✗	25.00
e010-twin	twin	36	weighted_tree	9	–	–	**0.776**	[0.748, 0.803]	✓	–
e011	HW	24	bfs_tree	7	0.893	0.790	**0.842**	[0.720, 0.972]	✓	18.00
e011-twin	twin	24	bfs_tree	7	–	–	**0.845**	[0.822, 0.867]	✓	–
e012	HW	28	weighted_tree	9	0.593	0.566	**0.580**	[0.444, 0.703]	✗	20.00
e012-twin	twin	28	weighted_tree	9	–	–	**0.815**	[0.792, 0.837]	✓	–

5 DISCUSSION AND LIMITATIONS

The lower bound of the reported fidelity record exceeds 1/2 by 0.020, which represents a narrow margin. This confidence interval accounts solely for shot noise and does not incorporate potential drift occurring between the 83 circuits of the experiment or across different experimental runs. A significant consideration is the selection bias introduced by the autonomous agent's methodology, which involved choosing configurations to execute and identifying a record result. Specifically, 6 hardware experiments were conducted within four qubits of the record size. Reporting the most favorable outcome among multiple noisy estimates inherently biases the reported fidelity upward. No correction for this selection effect was applied. Consequently, a pre-registered confirmation experiment, replicating the identical specification and ideally employing an increased number of shots, is necessary before the N = 40 result can be considered definitively established.

Readout-error mitigation was enabled in 100% of experiments. This technique reweights measured populations and parities using the inverse of a per-qubit confusion model, implying that the certified fidelity is a property of the mitigated estimator rather than of raw counts. The underlying model assumes readout errors are uncorrelated across qubits and free of crosstalk. Furthermore, the error rates are estimated from only two calibration circuits, and the shot noise inherent in these calibrations is not propagated into the bootstrap interval, which suggests the interval is somewhat too narrow. A conservative interpretation of the result would necessitate that the unmitigated fidelity also exceeds 1/2, a condition not separately established.

The digital twin simulation consistently overestimated hardware fidelity at every matched size, with discrepancies ranging from 0.02 to 0.40. This noise model is constructed from a single calibration snapshot and omits factors such as crosstalk, leakage, non-Markovian dephasing, and temporal drift. Therefore, the observed gap is expected to increase with N and with circuit duration. Since the twin predictions were finalized prior to the execution of the hardware jobs, the magnitude of this discrepancy provides a direct, unbiased measure of the portion of the device's error budget not captured by the model.

The experimental run spanned 6.6 hours, relying on a single calibration snapshot, during which device parameters may drift. While a fidelity exceeding 1/2 certifies genuine multipartite entanglement, it does not fully characterize the quantum state. Additionally, the witness relies on the assumption that the measurements performed are the intended ones. Finally, the agent's search strategy was brief and greedy, exploring only a limited portion of the space of roots, layouts, and settings. This suggests that larger certified states on this device remain plausible.

6 REPRODUCIBILITY

Run ID	`20261107-220808-qpu`
Run mode / backend	qpu / ibm_fez
Run start / end	2026-11-08 03:08 UTC / 2026-11-08 09:45 UTC
Calibration	live backend query, 2026-11-08T03:07:49+00:00
Budget used	322.0 / 480 QPU s; 16 / 20 LLM calls
Agent model	gemini:gemini-2.5-flash
Software	qiskit 2.6.2, qiskit-ibm-runtime 0.51.0, qiskit-aer 0.18.2, numpy 2.5.4

Hardware job IDs:

e001: `dbnuf68ou4as73b57070`; e002: `dbnuf6oqtr0c73admn4g`; e003: `dbnuf7dt7fqc73ak8im0`; e004: `dbnuf7oqtr0c73admn7g`; e005: `dbnuf8dt7fqc73ak8ing`; e006: `dbnuf8tt7fqc73ak8ip0`; e007: `dbnuf9gou4as73b570fg`; e008: `dbnufa8qtr0c73admne0`; e009: `dbo4738ou4as73b5fts0`; e010: `dbo473tt7fqc73akhc70`; e011: `dbo47444148s7391u70g`; e012: `dbo474k4148s7391u71g`

Every number in this paper is computed from `runs/20261107-220808-qpu/` (`calibration.json`, `events.jsonl`, `experiments/*.json`) and can be regenerated with `uv run python -m scripts.write_paper runs/20261107-220808-qpu`.

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